

**SIXTH SEMESTER (CUCBCSS—UG) DEGREE [SPECIAL] EXAMINATION
MARCH 2021**

Mathematics

MAT 6B 13 (E02)—LINEAR PROGRAMMING

Time : Three Hours

Maximum : 80 Marks

Section A

Answer all questions.

Each question carries 1 mark.

1. What is meant by a Polyhedral Convex set ?
2. State graphical solution algorithm for an LPP.
3. Write down the Mathematical form of a general LPP.
4. What are the slack and surplus variables ?
5. Define a basic solution of a system of m linearly independent equations with n unknowns.
6. State Minimax theorem.
7. Write down the dual of the following LPP :
Maximize $z = 100x_1 + 15x_2$
subject to the constraints :
$$x_1 + 4x_2 \leq 20$$
$$4x_1 + x_2 \leq 35$$
$$x_1, x_2 \geq 0.$$
8. What is the transportation problem ?
9. Define non-degenerate basic feasible solution.
10. What is an unbalanced transportation problem ? How can you convert it into a balanced transportation problem ?
11. Write down the mathematical formulation of an assignment problem.
12. What is the number of basic variables in a balanced transportation problem with m origins and n destinations.

(12 × 1 = 12 marks)

Turn over

Section B

Answer at least eight questions.

Each question carries 3 marks.

All questions can be attended.

Overall Ceiling 24.

13. A paper mill produces two grades of paper namely X and Y . Owing to raw material restrictions, it cannot produce more than 400 tons of grade X and 300 tons of grade Y in a week. There are 160 production hours in a week. It requires 0.2 and 0.4 hours to produce a ton of products X and Y, respectively with corresponding profit of Rs. 200 and Rs. 500 per ton. Formulate the above problem as an LPP to maximize the profit and find the optimum product mix.
14. Show that the set $S = \{(x_1, x_2) : 3x_1^2 + 2x_2^2 \leq 6\}$ is convex.
15. Write the characteristics of the standard form of Linear Programming Problem.
16. Write down the following LPP in standard form :
 Maximize $z = 3x_1 + 2x_2 + 5x_3$
 subject to the constraints :
 $2x_1 - 3x_2 \leq 3$
 $x_1 + 2x_2 + 3x_3 \geq 5$
 $3x_1 + 2x_3 \leq 2$
 $x_1, x_2, x_3 \geq 0.$
17. State the simplex algorithm for the computation of an optimum solution of LPP.
18. Explain two phase simplex method.
19. Discuss the relationship between primal and its dual.
20. Prove that every loop has an even number of cells.
21. Write all the steps for Vogel's Approximation methods of solving a transportation problem.
22. Explain degeneracy in a transportation problem.
23. In an assignment problem, if we add or subtract a constant to every element of a row or column of the cost matrix, then prove that an assignment plan which minimizes the total cost for the new matrix, also minimizes the total cost for the original cost matrix.
24. Describe a method of drawing minimum number of lines in the context of assignment problem.

(8 × 3 = 24 marks)

Section C

Answer at least five questions.

Each question carries 6 marks.

All questions can be attended.

Overall Ceiling 30.

25. Let A be an $m \times n$ matrix, and b an m- vector then show that $\{x \in \mathbb{R}^n : Ax \leq b\}$ is a convex set.

26. Solve the following LPP by graphical method :

Minimize $Z = 20x_1 + 10x_2$
subject to the constraints :

$$x_1 + 2x_2 \leq 40$$

$$3x_1 + x_2 \geq 30$$

$$4x_1 + 3x_2 \geq 60$$

$$x_1, x_2 \geq 0.$$

27. Find the degenerate basic feasible solutions of the system :

$$2x_1 + 6x_2 + 2x_3 + x_4 = 3$$

$$6x_1 + 4x_2 + 4x_3 + 6x_4 = 2$$

$$x_i \geq 0, i = 1, 2, 3, 4.$$

28. State and prove fundamental theorem of linear programming.

29. Use simplex method to solve the following LPP :

$$\text{Maximize } z = 3x_1 + 2x_2$$

subject to the constraints :

$$x_1 + x_2 \leq 4$$

$$x_1 - x_2 \leq 2$$

$$x_1, x_2 \geq 0.$$

30. Use penalty method to

$$\text{Maximize } z = 3x_1 + 2x_2 + 3x_3$$

subject to the constraints :

$$2x_1 + x_2 + x_3 \leq 2$$

$$3x_1 + 4x_2 + 2x_3 \geq 8$$

$$x_1, x_2, x_3 \geq 0.$$

31. Solve the following assignment problem in order to minimize the total cost. The cost matrix given below gives the assignment cost when different operators are assigned to various machines :

Machines	Operators				
	I	II	III	IV	V
A	30	25	33	35	36
B	23	29	38	23	26
C	30	27	22	22	22
D	25	31	29	27	32
E	27	29	30	24	32

Turn over

32. Prove that the necessary and sufficient condition for the existence of a feasible solution to the transportation problem is that total demand is equal to the total supply.
33. Determine an initial feasible solution to the following transportation problem using the north-west corner rule :

	D ₁	D ₂	D ₃	D ₄	Supply
O ₁	6	4	1	5	14
O ₂	8	9	2	7	16
O ₃	4	3	6	2	5
Required	6	10	15	4	35

(5 × 6 = 30 marks)

Section D

Answer at least **one** question.

The question carries 14 marks.

34. Let $A \subseteq \mathbb{R}^n$ be any set. Prove that $\langle A \rangle$, the convex hull of A, is the set of all finite convex combinations of vectors in A.
35. Solve the LPP by simplex method :

Minimize $z = x_1 - 3x_2 + 2x_3$

subject to the constraints :

$$3x_1 - x_2 + 2x_3 \leq 7$$

$$-2x_1 + 4x_2 \leq 12$$

$$-4x_1 + 3x_2 + 8x_3 \leq 10$$

$$x_1, x_2, x_3 \geq 0.$$

36. Find the optimal solution of the following transportation problem whose cost matrix is given as under :

	D ₁	D ₂	D ₃	D ₄	Supply
O ₁	2	2	2	1	3
O ₂	10	8	5	4	7
O ₃	7	6	6	8	5
Demand	4	3	4	4	15

(1 × 14 = 14 marks)